

380 — Sample Midterm

Answers and some notes. I typed this up quickly; let me know if you find any mistakes.

- This exam consists of five questions.
- Each question is worth 10 points.
- The exam is worth 40 points.
- Your exam score is the sum of your four highest question scores.
- Calculators etc. are not allowed.

1. Let $\Sigma = \{0, 1\}$. In this question, every string over Σ is viewed as a binary natural number. Leading 0s are allowed. For example, 00100 is viewed as the number 4. Λ is viewed as the number 0. Look at the following list of six languages.

- $L_1 = \{x \in \Sigma^* \mid x \geq 16\}$.
- $L_2 = \{x \in \Sigma^* \mid x \text{ is a square and } x < 1000\}$.
- $L_3 = \{x \in \Sigma^* \mid x \text{ is not divisible by } 3\}$.
- $L_4 = \{x \in \Sigma^* \mid x \text{ is divisible by } 666\}$.
- $L_5 = \{xy \mid x, y \in \Sigma^* \text{ and } x * y = 12\}$.
- $L_6 = \{xy \mid x, y \in \Sigma^*, |x| = |y|, \text{ and } x \text{ and } y \text{ represent the same number}\}$.

- (a) Which of the languages from the list are regular?

L_1, L_2, L_3, L_4 , and L_5 .

- (b) Which of the languages from the list contain Λ ?

L_2, L_4 , and L_6 .

- (c) Draw the transition diagram of a FA that accepts one of the languages from the list *or* give a regular expression that represents one of the languages from the list. Also state which language you chose.

Different answers are possible. You want to pick a case that is easy and that you are sure about. And you should do just one case!

- Regular expression for L_1 :

$$0^*1(0+1)^*(0+1)^4$$

The FA for L_1 is also easy.

- *We did an FA for “divisible by 3” on one of the homeworks. Switching the accepting and rejecting states will give you an FA for L_3 .*
- *L_6 is pretty easy as well. Regular expression:*

$$0^*10^*1100 + 0^*100^*110 + 0^*110^*100 + 0^*1000^*11 + 0^*1100^*10 + 0^*11000^*1$$

- *The regular expression for L_4 is easy, but long. You need to “+” together all squares less than 1000 in binary and precede that by 0^* , since leading 0s are allowed.*

2. Let M be the following NFA- Λ . $M = (Q, \Sigma, q_0, A, \delta)$ such that $Q = \{1, 2, 3, 4\}$, $\Sigma = \{a, b\}$, $q_0 = 1$, $A = \{3\}$, and δ is given by the following transition table:

q	$\delta(q, \Lambda)$	$\delta(q, a)$	$\delta(q, b)$
1	$\{2\}$	$\emptyset$	$\emptyset$
2	$\{3\}$	$\emptyset$	$\{2\}$
3	$\emptyset$	$\{3, 4\}$	$\{1\}$
4	$\emptyset$	$\{2\}$	$\emptyset$

- (a) Draw the transition diagram of M .

I assume that all of you are capable of doing this.

- (b) The book describes how to transform an NFA- Λ into an equivalent NFA. Draw the transition diagram of the equivalent NFA that is produced by applying this algorithm on M . Do *not* simplify the NFA.

I will give the quintuple definition; you can convert this to the transition diagram.

$M = (Q, \Sigma, q_0, A, \delta)$ such that $Q = \{1, 2, 3, 4\}$, $\Sigma = \{a, b\}$, $q_0 = 1$, $A = \{1, 3\}$, and δ is given by the following transition table:

q	$\delta(q, a)$	$\delta(q, b)$
1	$\{3, 4\}$	$\{1, 2, 3\}$
2	$\{3, 4\}$	$\{1, 2, 3\}$
3	$\{3, 4\}$	$\{1, 2, 3\}$
4	$\{2, 3\}$	$\emptyset$

- (c) Give a regular expression that is equivalent to $L(M)$. Make your expression as simple as possible.

Answer: $(a + b)^$*

3. Let $\Sigma = \{a, b\}$. Let *double* be the function from Σ^* to Σ^* that doubles each character in a string. For example, $\text{double}(\text{baaba}) = \text{bbaaaabbaa}$.

(a) What is $\text{double}(\text{aabaa})$?

aaaabbbaaaa

(b) What is $\text{double}(\Lambda)$?

Λ

(c) Suppose $x \in \{a, b\}^*$ and the length of x is k . What is the length of $\text{double}(x)$?

$2k$

(d) Give a recursive definition of function *double* from Σ^* to Σ^* .

- $\text{double}(\Lambda) = \Lambda$.
- For all $x \in \Sigma^*$ and $\sigma \in \Sigma$, $\text{double}(x\sigma) = \text{double}(x)\sigma\sigma$.

For L a language over Σ , define $\text{double}(L)$ as follows:

$$\text{double}(L) = \{\text{double}(x) \mid x \in L\}.$$

(e) Let A be the language $\{ab, bbb, baba\}$. What is $\text{double}(A)$?

$\{aabb, bbbbbb, bbaabbaa\}$

(f) List all languages B over Σ that have the property that $\text{double}(B) = B$.

$\emptyset$ and $\{\Lambda\}$

(g) For each of the following statements, circle the right answer.

- i. If L is a regular language over Σ , then $\text{double}(L)$ is regular. *True*
- ii. If L is a finite language over Σ , then $\text{double}(L)$ is finite. *True*
- iii. If $L \subseteq \Sigma^*$ is not regular, then $\text{double}(L)$ is not regular. *True*
- iv. If L_1 and L_2 are regular languages over Σ , then $L_1 \cup L_2$ is regular. *True*
- v. If L_1 and L_2 are finite languages over Σ , then $L_1 \cup L_2$ is finite. *True*
- vi. If $L_1 \subseteq \Sigma^*$ and $L_2 \subseteq \Sigma^*$ are not regular, then $L_1 \cup L_2$ is not regular. *False*

4. Let $L = \{a^i b^k a^\ell \mid \ell > i + k\}$.

(a) List all strings in L of length 7.

$a^7, ba^6, aba^5, bba^5, aaba^4, abba^4, b^3a^4$

(b) Use the Pumping Lemma for Regular Languages to prove that L is not regular.

Suppose for a contradiction that L is regular. Let n be the integer from the pumping lemma.

Let $x = b^n a^{n+1}$. Then $x \in L$ and $|x| \geq n$. By the pumping lemma, there exist strings u, v, w such that $x = uvw$, $|uv| \leq n$, $|v| > 0$, and $uv^m w \in L$ for all $m \geq 0$.

In particular, $uv^2w \in L$. (Choose $m = 2$.)

Since $|uv| \leq n$, $uv^2w = b^{n+|v|}a^{n+1}$. Since $|v| > 0$, $n+1 \neq n+|v|$. It follows that $uv^2w \notin L$. This is a contradiction.

It follows that the assumption that L is regular is wrong.

So, we have shown that L is not regular.

5. This question is about the subset construction. In this question, take $\Sigma = \{a, b, c\}$.

- (a) If you literally apply the subset construction to an NFA with k states, how many states will the corresponding FA have?

2^k

- (b) Give a simple example of a minimal NFA such that the FA obtained by literally applying the subset construction is *not* minimal. (A minimal NFA is an NFA such that no equivalent NFA has fewer states.) For your answer, draw the transition diagram of the NFA, the transition diagram of the FA obtained by applying the subset construction, and briefly argue that the FA is not minimal.

Consider the NFA M_1 with one rejecting state and no transitions (you need to draw the transition diagram). Clearly, M_1 is minimal, since an NFA can not have fewer than one state. $L(M_1) = \emptyset$ and there exists a one-state FA that accepts $\emptyset$. But the subset construction gives a two-state FA (you need to draw the transition diagram).

- (c) Give a simple example of a minimal NFA such that the FA obtained by literally applying the subset construction is minimal. For your answer, draw the transition diagram of the NFA, the transition diagram of the FA obtained by applying the subset construction, and briefly argue that the FA is minimal.

Consider the NFA M_2 with one accepting state and no transitions (draw it). Clearly, M_2 is minimal, since an NFA can not have fewer than one state. $L(M_2) = \{\Lambda\}$. The two-state FA given by the subset construction (draw it) is minimal, since an FA that accepts Λ needs at least one accepting state, and an FA that does not accept Σ^ needs at least one rejecting state.*